

ANJOOM THAHIR ALIKKAM VEETIL
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EDUCATION

Master of Science (by Research) in Cognitive Science

Department of Cognitive Science, Indian Institute of Technology Kanpur, Kanpur, India

Jul. 2018 – Aug. 2021 · degree conferred 28 Dec. 2021 · CPI 8.14/10

- Thesis: *Investigating the Differential effects of handedness continuum on the lateralization of low-level and high-level cognitive functions* (May 2019 – Aug. 2021). Supervisor: Dr. Ark Verma.
- Designed and conducted four behavioural experiments using the divided visual half-field paradigm to test the role of the handedness continuum in lateralized visual attention, recruiting 104 participants across two institutions and analysing data from 102.
- Showed that the handedness continuum modulates the degree and direction of lateralization in global–local processing (RT $r(96) = -0.41$, $p < .001$; $F(2,95) = 10.669$, $p < .001$, $\eta^2_p = 0.183$) and in face processing (RT $r(96) = -0.32$, $p = .001$), but not in orientation or spatial-frequency processing; the word-processing effect was weaker (RT $r(98) = -0.23$, $p = .022$) and did not survive a categorical group comparison.
- Identified low-level visual features as a potential confound in lateralization studies, and argued that action representation and spatial attention are inextricably interwoven.
- Research output: first-author publication in the *Proceedings of the Annual Meeting of the Cognitive Science Society* (2023); conference talk at ESCOP-2023; posters at CogSci 2023 and ACCS-2022.

Integrated BS-MS in Biology (Five-year BS-MS Dual Degree Programme)

Indian Institute of Science Education and Research Mohali, Mohali, India

Aug. 2013 – May 2018 · both degrees conferred 20 May 2018 · CPI 7.8/10

- Thesis: *Rodent-specific rearrangement between hyaluronidase and chemokine receptor gene clusters and its implication in cancer resistance and aging* (May 2017 – Jun. 2018). Supervisor: Dr. Kuljeet Singh Sandhu.
- Tested a published claim that reduced genomic distance between oncogenes and tumour suppressor genes lowers cancer incidence; when the initial observations contradicted the hypothesis, extended the analysis rather than abandoning it.
- Discovered a massive genomic rearrangement in the naked mole-rat genome that splits the tumour suppressor gene cluster relative to humans, shedding light on that species' exceptional cancer resistance.
- Demonstrated by comparative analysis across mammals that this convergent rearrangement is confined to two rodent suborders that also show reduced oncogenesis; managed and analysed large genomic datasets in Bash and R.
- Research output: co-first-author publication in *npj Aging and Mechanisms of Disease* (2021).

Higher Secondary Certificate (Class XII), Science stream

Board of Higher Secondary Examination, Government of Kerala; Government Higher Secondary School, Medical College Campus, Calicut, Kerala, India

2011 – Mar. 2013 · result published 08 May 2013

- Subject marks as printed on the certificate (each out of 200): Malayalam 197 (A+), Physics 188 (A+), Biology 183 (A+), Mathematics 183 (A+), Chemistry 178 (A), English 176 (A).

PUBLICATIONS

Peer-reviewed journal article

1. Jain, Y., Chandradoss, K. R., Alikkam Veetil, A., Bhattacharya, J., Lal, M., Bagadia, M., Singh, H., & Sandhu, K. S. (2021). Convergent evolution of a genomic rearrangement may explain cancer resistance in hystrico- and sciuromorpha rodents. *npj Aging and Mechanisms of Disease*, 7*(20), 1–15. <https://doi.org/10.1038/s41514-021-00072-9>

* Joint first authors.

Peer-reviewed conference proceedings

2. Alikkam Veetil, A., Roy, N., & Verma, A. (2023). Handedness modulates spatial attention – Insights from individual variations in lateralization of cognitive functions. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 45, 2705–2712. <https://escholarship.org/uc/item/0w62x0nt>

- In M. Goldwater, F. K. Anggoro, B. K. Hayes, & D. C. Ong (Eds.), *Proceedings of the 45th Annual Conference of the Cognitive Science Society*. Open access, CC BY 4.0.

Manuscripts in preparation — ready for submission (2026)

3. **Alikkam Veetil, A., & Verma, A.** (2026a, in preparation). *Individual variation at the perception–action interface: Signed handedness predicts lateralised response initiation and inhibition in hierarchical visual attention.*

4. **Alikkam Veetil, A., & Verma, A.** (2026b, in preparation). *Continuous handedness grades visual-field asymmetry across the visual hierarchy.* Intended for *Journal of Experimental Psychology: General, Cortex or Neuropsychologia*.

5. **Alikkam Veetil, A., & Verma, A.** (2026c, in preparation). *The Graded Attention–Action Lateralisation Axis: A two-level developmental framework for individual variation in the hemispheric organisation of perception and action.* Intended for *Psychological Review, Neuroscience & Biobehavioral Reviews* or *Perspectives on Psychological Science*.

A confirmatory preregistration for the programme (core plus four appendices) has also been prepared.

CONFERENCE PRESENTATIONS

Talks

- **Alikkam Veetil, A.** (2024, November). *Spatial correlation between disorder-associated alterations of brain structural asymmetry and various neurobiological features.* Seminar talk presented at the Imaging Genomics Group Meeting, Language and Genetics Department, Max Planck Institute for Psycholinguistics, Nijmegen, The Netherlands.
- **Alikkam Veetil, A., Roy, N., & Verma, A.** (2023, September). *Handedness continuum captures individual variations in lateralization of cognitive functions.* Conference talk presented at the 23rd Conference of the European Society for Cognitive Psychology (ESCP-2023), University of Porto, Portugal.

Posters

- **Alikkam Veetil, A., Roy, N., & Verma, A.** (2023, July). *Handedness modulates spatial attention: Insights from individual variations in lateralization of cognitive functions.* Poster presented at the 45th Conference of the Cognitive Science Society (CogSci 2023), University of Sydney, Australia.
- **Alikkam Veetil, A., Roy, N., & Verma, A.** (2022, December). *Does the handedness continuum capture individual variations in the lateralization of cognitive functions?* Poster presented at the 9th Annual Conference of Cognitive Science (ACCS-2022), Indian Institute of Technology Delhi, India.

RESEARCH EXPERIENCE

Master's thesis research at IIT Kanpur (May 2019 – Aug. 2021) and IISER Mohali (May 2017 – Jun. 2018) is detailed under EDUCATION.

Independent Researcher · Handedness and the lateralization of visuospatial attention

Feb. 2026 – present · Self-directed, in collaboration with Dr. Ark Verma, Department of Cognitive Science, IIT Kanpur

- Returned to the behavioural dataset underlying the CogSci 2023 proceedings paper, which reported preliminary results only, and carried the analysis through to completion.
- Produced a three-paper programme — two empirical papers and one theoretical framework paper — together with a confirmatory preregistration, all co-authored with Dr. Ark Verma and ready for submission.
- Fitted a hierarchical Bayesian drift-diffusion model to 23,273 trials in R, estimating all four parameters jointly, and localised the visual-field × handedness interaction to nondecision time; supported by LOO model comparison, parameter recovery, prior-sensitivity analysis and convergence diagnostics.
- Applied permutation-null testing, specification-curve analysis, equivalence testing and cross-site heterogeneity statistics (I^2), reporting site-unstable indices as heterogeneous rather than as effects.
- Built a reproducible release pipeline with automated pre-send audits, cross-paper consistency checks and versioned release packages containing manuscripts, supplements, figures, participant-level data and results tables.
- Used AI tools for analysis support — writing and debugging analysis code, statistical reasoning, literature synthesis — and for manuscript editing, developing the theoretical framework and interpretation independently.
- Set the scope, sequence and submission targets for the project without institutional support or funding.

Talent+ Research Intern · Max Planck Institute for Psycholinguistics, Nijmegen, The Netherlands

Sep. 2024 – Dec. 2024 · Supervisors: Dr. Clyde Francks and Dr. Meng-Yun Wang, Imaging Genomics group, Language and Genetics Department

- Project: *Spatial correlation between disorder-associated alterations of brain structural asymmetry and various neurobiological features.*
- Applied multivariate statistics, including principal component analysis and canonical correlation analysis, to disorder-associated measures of brain structural asymmetry.
- Processed structural MRI data with FreeSurfer pipelines to extract cortical and subcortical measures.

- Presented the project as a seminar talk at the Imaging Genomics Group Meeting (Nov. 2024).
- Placement funded by the Max Planck Institute, which met all travel and accommodation costs.

Computational Neuroscience course project · Neuromatch Academy (remote)

Jul. 2023

- Implemented models to decode neural signals for visual sequence detection in early visual cortex, strengthening both coding and conceptual skills.
- Completed the interactive track **and the course project** of Computational Neuroscience (certificate dated 28 Jul. 2023).

Research Assistant · Department of Cognitive Science, Indian Institute of Technology Kanpur, Kanpur, India

Jan. 2019 – Apr. 2019 · Supervisor: Dr. Ark Verma

- Project: *Embodiment in lateralization*.
- Designed human behavioural experiments and conducted pilot studies on embodied action representation.

Summer Research Scholar · Department of Biological Science, Indian Institute of Science Education and Research Mohali, Mohali, India

Mar. 2017 – May 2017 · Supervisor: Dr. Kavita Babu

- Project: *Chemotaxis in C. elegans*.
- Designed and conducted chemotaxis experiments with *C. elegans*.

Summer Research Scholar · Department of Biological Science, Birla Institute of Technology and Science (BITS) Pilani, K. K. Birla Goa Campus, Goa, India

May 2016 – Jul. 2016 · Supervisor: Dr. Anasuya Ganguly

- Maintained the HeLa cancer cell line; performed cloning, gene expression analysis and protein extraction.

TEACHING EXPERIENCE

Teaching Assistant · Department of Cognitive Science, Indian Institute of Technology Kanpur, Kanpur, India

Aug. 2019 – Dec. 2019

- Assisted in the delivery of *Introduction to Cognitive Science*. Course Instructor: Dr. Ark Verma.

Teaching Assistant · Department of Biological Science, Indian Institute of Science Education and Research Mohali, Mohali, India

Aug. 2017 – May 2018

- Assisted in *Biology Lab II* and *Biology Lab III*.
- Guided first- and second-year students in developing experimental designs for laboratory projects.

AWARDS, FELLOWSHIPS AND HONOURS

- **Talent+ Research Internship**, Max Planck Institute for Psycholinguistics, Nijmegen — Sep. 2024 – Dec. 2024. Fully funded placement; all travel and accommodation costs met by the Institute.
- **Diversity and Inclusion Conference Award, CogSci 2023, USD 1,000** — May 2023. One of ten awarded by the Cognitive Science Society, to the first author of an accepted paper, to promote conference participation by graduate students who bring diversity to the society.
- **Institute fellowship for postgraduate students**, Indian Institute of Technology Kanpur — Aug. 2018 – Aug. 2020.
- **INSPIRE fellowship**, Department of Science and Technology, Government of India — Aug. 2013 – May 2018. Instituted to promote basic science research.

COMPETITIVE EXAMINATIONS AND ACADEMIC DISTINCTIONS

- **All India Rank 74 (Lectureship)**, Life Science paper, National Eligibility Test (NET) — Dec. 2018.
- **All India Rank 1868**, Life Science paper, Graduate Aptitude Test in Engineering (GATE) — Dec. 2018.
- **Placed in the top 450 nationally**, IISER Aptitude Test (IAT) — Jul. 2013.
- Participant, winter science camp organised by NIT Calicut — 2011.

TECHNICAL AND RESEARCH SKILLS

- **Programming and data:** Python — NumPy, pandas, SciPy, Matplotlib, scikit-learn; used for behavioural data pipelines, psychometric analysis and figure generation. R — brms and Stan for hierarchical Bayesian

models, lme4/lmerTest and emmeans for mixed-effects analysis, loo for model comparison, posterior and tidybayes for posterior summaries, dplyr/tidyr and ggplot2. Bash (Intermediate), Perl (Intermediate).

- **Bayesian modelling:** hierarchical Bayesian regression in brms/Stan (Hamiltonian Monte Carlo, NUTS); **hierarchical drift-diffusion modelling using the Wiener likelihood**, with all four parameters estimated jointly on 23,273 trials; Bernoulli and binomial accuracy models; leave-one-out cross-validation and model comparison ('loo'); parameter recovery; prior-sensitivity analysis; convergence diagnostics (R-hat, effective sample size, tree-depth and divergence checks); posterior summarisation and visualisation; parallelised sampling.
- **Frequentist and psychometric analysis:** linear and generalised linear mixed-effects models (lme4, lmerTest); estimated marginal means and simple-slopes decomposition (emmeans); factor analysis (scikit-learn); principal component analysis; canonical correlation analysis; inferential statistics to APA standard (correlation with df, one-way ANOVA with partial η^2 , Cohen's d, Holm-corrected post-hoc tests, Kolmogorov-Smirnov tests, outlier control).
- **Inference robustness:** permutation-null construction and testing; specification-curve analysis; equivalence testing; cross-site heterogeneity statistics (I^2) with random-effects site models; sensitivity and site-composition sweeps; external-dataset replication checks.
- **Open and reproducible research:** preregistration of a confirmatory programme; reproducible analysis pipelines; automated pre-send audit suites and cross-document consistency checking; versioned release packaging of manuscripts, supplements, data, figures and results with manifests.
- **Neuroimaging:** structural MRI morphometry — extraction of cortical and subcortical measures with FreeSurfer (Basic); EEG (Basic).
- **Experimental design and instrumentation:** divided visual half-field paradigm; psychophysical stimulus design and millisecond-level timing control; Go/No-Go and bimanual response protocols; spatial-frequency grating stimuli; hierarchical (Navon) global-local stimuli; face matching using the Radboud Faces Database; psycholinguistically normed word stimuli controlled for concreteness, valence, arousal and frequency; modified Edinburgh Handedness Inventory scored as a continuum, with item-level psychometric analysis; LexTALE.
- **Specialised software:** PsychoPy (Advanced — Builder plus custom code components), JASP (Intermediate), Overleaf (Intermediate), FreeSurfer (Basic).
- **Research operations:** multi-site participant recruitment and management, including quota sampling for a uniform handedness distribution across two institutions (n = 62 and n = 42).
- **Molecular biology techniques:** cloning, immunohistochemistry, PCR, RT-PCR, SDS-PAGE, Western blot.
- **Molecular imaging techniques:** fluorescence microscopy, confocal microscopy.
- **Comparative genomics:** cross-species gene-distance analysis, synteny and rearrangement detection, large-dataset handling in Bash and R.

RELEVANT COURSEWORK

Selected coursework; grades as printed on the official transcripts (A = 10, B = 8, C = 6, S = Satisfactory). Full transcripts available on request.

Master of Science (by Research) in Cognitive Science, IIT Kanpur

- Cognitive science core: Foundations of Cognitive Science (C), Methods and Tools in Cognitive Science (B), Logic and Cognitive Science (B), Computational Tools for Cognitive Science (C), Seminar on Cognitive Science (B), Project (A).
- Psychology and neuroscience: Human Cognitive Processes (A), Neurobiology (A), Topics in Cognitive Neuroscience (S), Psychology of Language (B).
- Research: MS Research Thesis, 90 credits (S).

Integrated BS-MS in Biology, IISER Mohali

- Quantitative, computational and interdisciplinary: Theory of Computation (A), Introduction to Computers (A), Network Science (B), Computational Biology (B), Experimental Design and Hypothesis Testing (B), Bioinformatics Lab (C), Probability and Statistics (C).
- Molecular and cell biology: Molecular Methods in Biology Lab (A), Advanced Cell Biology (B), Gene Expression and Development (B), Structure and Function of Genomes (C), RNA Biology (C), Cancer Cell Biology (C), Cell and Molecular Basis of Immune Response (C).
- Organismal, evolutionary and behavioural biology: Advanced Evolutionary Biology (B), Behaviour and Ecology (B), Animals: Form and Function (B), Advanced Developmental Biology (C), Genetics and Evolution (C), Microbial Physiology and Microbial Genetics (C).
- Laboratory training: Biology Laboratory I (A), Biology Laboratory II (A), Biology Laboratory IV (A), Development and Physiology Lab (A), Biophysics and Spectroscopic Tools Lab (A), Biology Lab III (B).

- Humanities: History of Science (B), From Plassey to Partition: A History of Modern India (B), The Social History of Science in Modern India, 1780–1950 (B), Language Skills (B).
- Physics, chemistry and mathematics: four semesters each of physics and chemistry laboratory work (grades A and B), Electromagnetism (B), Spectroscopic and Other Physical Methods (B), Advanced Optics and Spectroscopy Lab (B), Analysis in One Variable (C), Symmetry (C), Curves and Surfaces (C).
- Research: Thesis Research (16 credits, A); Thesis Research II (16 credits, A).

WORKSHOPS, SUMMER SCHOOLS AND ADDITIONAL TRAINING

- MEDAL workshop, "Open Science: Why and how to do preregistration?", Max Planck Institute for Psycholinguistics — Nov. 2024
- Workshop on Statistics, Max Planck Institute for Psycholinguistics — Nov. 2024
- Workshop on Academic Writing, Max Planck Institute for Psycholinguistics — Nov. 2024
- Workshop on introduction to eye tracking and neuroimaging, Max Planck Institute for Psycholinguistics — Oct. 2024
- Workshop on Open Science practices, Max Planck Institute for Psycholinguistics — Oct. 2024
- Diversity, Equity and Inclusion workshop on "Unconscious Bias", Max Planck Institute for Psycholinguistics — Oct. 2024
- Symposium, "Speech and Language Disorders: Lessons from Genetics", Max Planck Institute for Psycholinguistics — Sep. 2024
- Climate science workshop organised by LUCA, KSSP — Dec. 2023
- Computational Neuroscience, Neuromatch Academy — Jul. 2023 (certificate: interactive track and course project)
- Neurocognitive Foundations of Mind 2022 (online conference) — Oct. 2022
- 8th North Sea Laterality Conference on Brain Asymmetry, University of Bergen (virtual) — Aug. 2022
- 11th Summer School of IMPRS NeuroComm, Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig (virtual) — Jun. 2022
- Workshop on Cognitive Neuroimaging, Centre of Behavioural and Cognitive Sciences, University of Allahabad — Jan. 2020
- 6th Annual Conference of Cognitive Science (ACCS-2019), BITS Pilani, K. K. Birla Goa Campus — Dec. 2019
- Winter School on Cognitive Modelling, Indian Institute of Technology Mandi — Feb. 2019
- GIAN course on Natural Language Processing, Indian Institute of Technology Kanpur — Dec. 2018
- 5th Annual Conference of Cognitive Science (ACCS-2018), Indian Institute of Technology Guwahati — Oct. 2018

ACADEMIC SERVICE AND OUTREACH

- Educational camps on sustainable living and permaculture for diverse groups — Urvi Earth Centre (Jan. 2025 – Sep. 2025) and Farmer's Share Permaculture and Craft Learning Center (Mar. 2024 – Jun. 2024). Curated the educational modules at Farmer's Share.

PROFESSIONAL EXPERIENCE OUTSIDE ACADEMIA

The conference presentations of Dec. 2022, Jul. 2023 and Sep. 2023, the 2023 CogSci proceedings publication and Neuromatch Academy were all completed during the period covered below.

Independent study — food-forest systems and tribal architecture · Dhaanyam, Attappadi, Kerala, India
Jan. 2026

- Studied food-forest design and traditional tribal architecture, extending the sustainable-systems work begun at Urvi Earth Centre.

Festival volunteer · Dharamshala International Film Festival, Himachal Pradesh, India
Late 2025

- Volunteered at the festival; completed two Vipassana meditation courses and travelled during the same period.

Volunteer, then Community and Sustainability Manager · Urvi Earth Centre, Munnar, Kerala, India
Jan. 2025 – Sep. 2025

- Spearheaded vernacular construction projects using low-carbon, earthen materials, prioritising climate-resilient and traditional building techniques.

- Implemented circular economy principles through audits of site-sourced and bio-based materials to eliminate construction waste.
- Facilitated mindfulness programmes (Sadhana) exploring how physical environments and internal practices shape well-being.

Operations Manager · Farmer's Share Permaculture and Craft Learning Center, Shornur, Kerala, India

Mar. 2024 – Jun. 2024

- Directed daily operations for an organic farming collective, systematising farm workflows and resource management.
- Curated educational modules on sustainable living and permaculture, leading immersive camps for diverse groups.
- Managed community infrastructure, coordinating agricultural activities with resident needs.

Managing Director · Growers Haven LLP, Pollachi, Tamil Nadu, India

Jul. 2022 – Mar. 2024

- Incorporated the limited liability company and managed its daily operations.
- Oversaw end-to-end operations, including procurement, processing, packaging and export logistics, coordinating teams and processes to meet delivery targets.

LANGUAGES

- English — Advanced. Medium of instruction throughout both degrees; certified by Medium-of-Instruction letters from IIT Kanpur (24 Nov. 2021) and IISER Mohali (17 May 2018). Class XII English: 176/200, grade A.
- Malayalam — Native. Class XII Malayalam: 197/200, grade A+.
- Hindi — Intermediate.
- Tamil — Intermediate.

REFEREES

Dr. Ark Verma — Master's thesis supervisor

Assistant Professor, Department of Cognitive Science, Indian Institute of Technology Kanpur, Kanpur, India
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Dr. Clyde Francks — Talent+ internship mentor

Associate Professor, Language and Genetics Department, Max Planck Institute for Psycholinguistics, Nijmegen, The Netherlands
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Dr. Kuljeet Singh Sandhu — Master's thesis supervisor

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